FlexiPath –
switch before the link fails

Keep a drone's control link, voice and video alive when it moves between Wi-Fi, 5G, satellite and wired networks. Switch before the signal fails.

TEAM TAROT CLUB

Al Ain University

EDGE GROUP CHALLENGE

Advanced Technology Pioneers 2026

LIVE PROTOTYPE

flexipath.vercel.app

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▶ **TRY THE LIVE PROTOTYPE**

flexipath.vercel.app

guided demo, simulator and the 2-minute video

Executive summary

FlexiPath keeps critical connections alive with little extra power

PROBLEM	In our model, a legacy stack leaves one network before it joins the next. A moving user loses the connection. When the control link (C2) of a drone is silent for more than 300 ms, the drone must stop and hover (our piloted-flight rule).
MECHANISM	FlexiPath finds a fading signal seconds early. It wakes the next network and tests it. For 0.4 s, it sends only the critical commands on both networks. Then it switches with no gap. An anchor relay keeps each connection open. Spare modems sleep at all other times to save power.
EVIDENCE	We run 160 paired simulated flights (8 scenarios, 5 seeds, 4 speeds) with a legacy stack and two existing methods. Reactive multipath (MPTCP/ATSSS) switches after failure. Always duplicate (PDCP duplication) sends twice. With our 300 ms rule and 60 vs 250 ms outage detection, FlexiPath keeps C2 on 100 % of flights. A legacy stack keeps it on 0 %. Reactive multipath fails in three scenarios. Always duplicate uses 2.1–2.6× the power.
DELIVERED	**4–7 points more reliability per extra watt for FlexiPath** versus 0.3–0.9 for always duplicate. A working prototype (interactive 3D digital twin with a guided demo), a deterministic simulation with a headless benchmark, and a documented model. We check its parameters against published data. All results are from simulation; we claim no live-network measurements.

The problem and why it matters

In our model, every network change costs a legacy stack 3.8 s of silence.

Dropped calls. Frozen streams.
300 ms of silence makes a drone hover.

Emergency response, industrial automation, connected transport and defence cannot wait.

Two drones, one route: Wi-Fi → 5G → satellite.

Why drones: we model a hover after 300 ms without C2 (our piloted-flight rule).

1. DOCK (WIRED)
Stable baseline mission prep

2. WI-FI 6
Local coverage on the move

3. 5G SA
Wide-area coverage at speed

4. LEO SATELLITE
Global reach beyond line of sight

CONTROL LINK (C2)
Command and control

PUSH-TO-TALK VOICE (REAL-TIME)
Clear, low-latency communications

CAMERA UPLOADS (BULK)
High-throughput data transfer

Existing mechanisms we analysed (criterion 01)

REACTIVE MULTIPATH

MPTCP backup subflow (RFC 8684) and 3GPP ATSSS active-standby.

Keeps a backup network on standby. It moves only after it confirms losses and the spare modem wakes up.

0.35 s
gap, more than our 300 ms C2 budget

1.20 W
low power consumption

ALWAYS DUPLICATE

MPTCP redundant scheduler and PDCP duplication.

Sends every important packet on two networks, all the time.

99-100 %
critical delivery (most reliable)

2.65-3.39 W
high power consumption and copies 49-100 % of protected bytes

Still fails when both paths fail together (a shared radio shadow).

COMMON SHORTCOMINGS

Neither looks ahead.

Ignores the fading signal.

Ignores the known route.

Neither adds protection only at the moment a link fades.

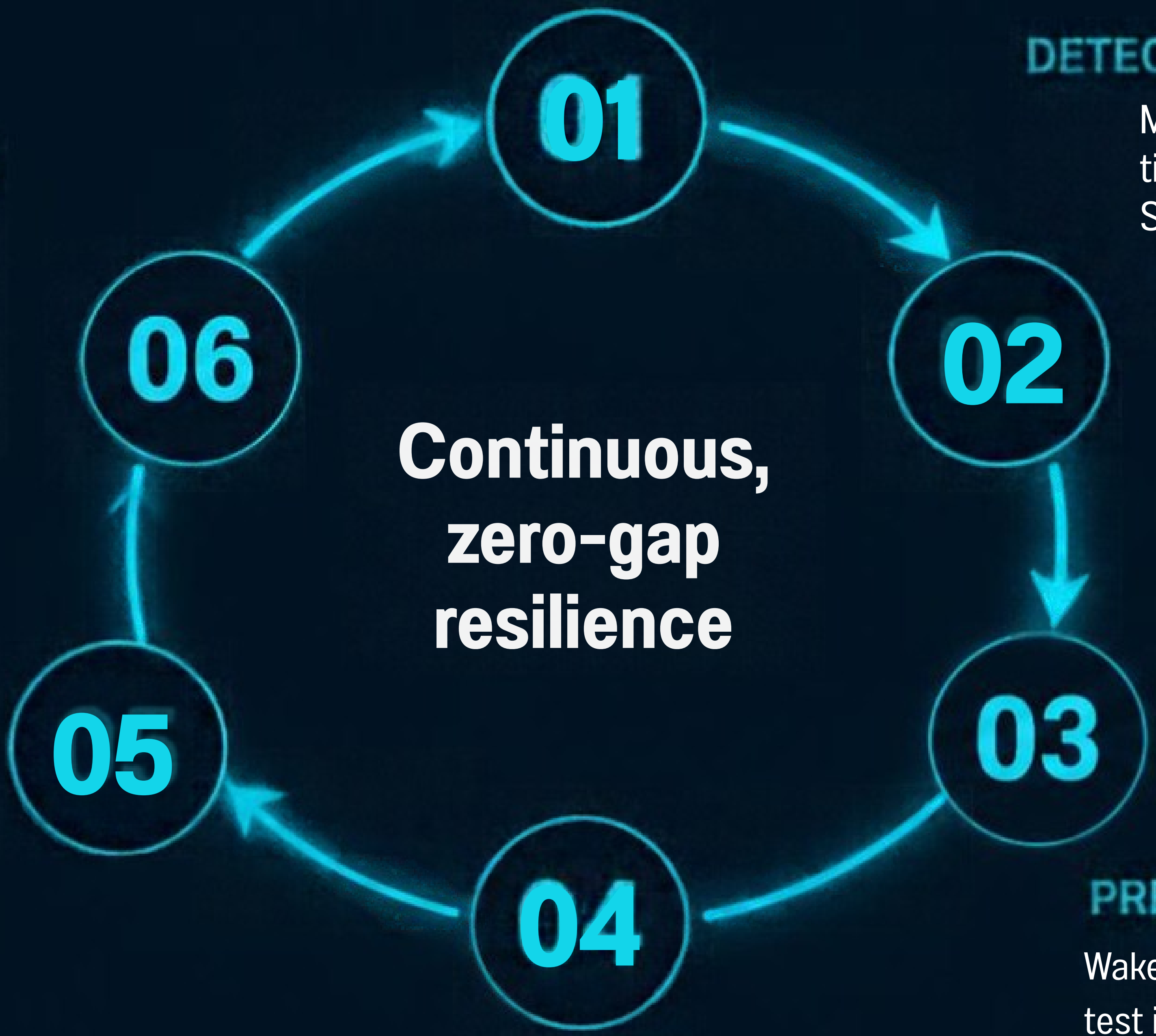
WHAT WE ADAPT

Keep the **anchored session** (from multipath).
Keep the **duplication** (from PDCP), but use it only in a **400 ms** burst around a predicted failure.
New in combination: **prediction**, **route knowledge** and **traffic classes**.

References: RFC 8684 MPTCP v1 (2020) · 3GPP TS 23.501 §5.32, ATSSS · 3GPP TS 38.323, PDCP duplication · 3GPP TS 22.125 §7.2, UAV C2 KPIs · Related: Elsight Halo, Peplink SpeedFusion, uAvionix muLTElink

The FlexiPath mechanism

(criterion 02)



Continuous, zero-gap resilience

DETECT

Measure the primary link ten times a second.
Spare modems sleep in DRX at 1.2 W and report signal once per cycle.

PREDICT

Extend the signal trend to find where packet loss reaches 15 %.
Watch from 8 s to failure; act under 3 s.
The coverage map warns 8 s before a known shadow.

PREPARE

Wake the best spare modem (1.8 W) and test it with small 64-byte heartbeats.
Check its delay and loss before it carries anything.

REPLICATE

For a 400 ms burst (3.4 W), send critical traffic on both paths. Add real-time traffic if its loss is over budget.
The receiver keeps the first copy of each sequence number and discards the late duplicate, so head-of-line delay is 0 ms.
Never copy bulk traffic; only pace it.

HONEST BEHAVIOUR

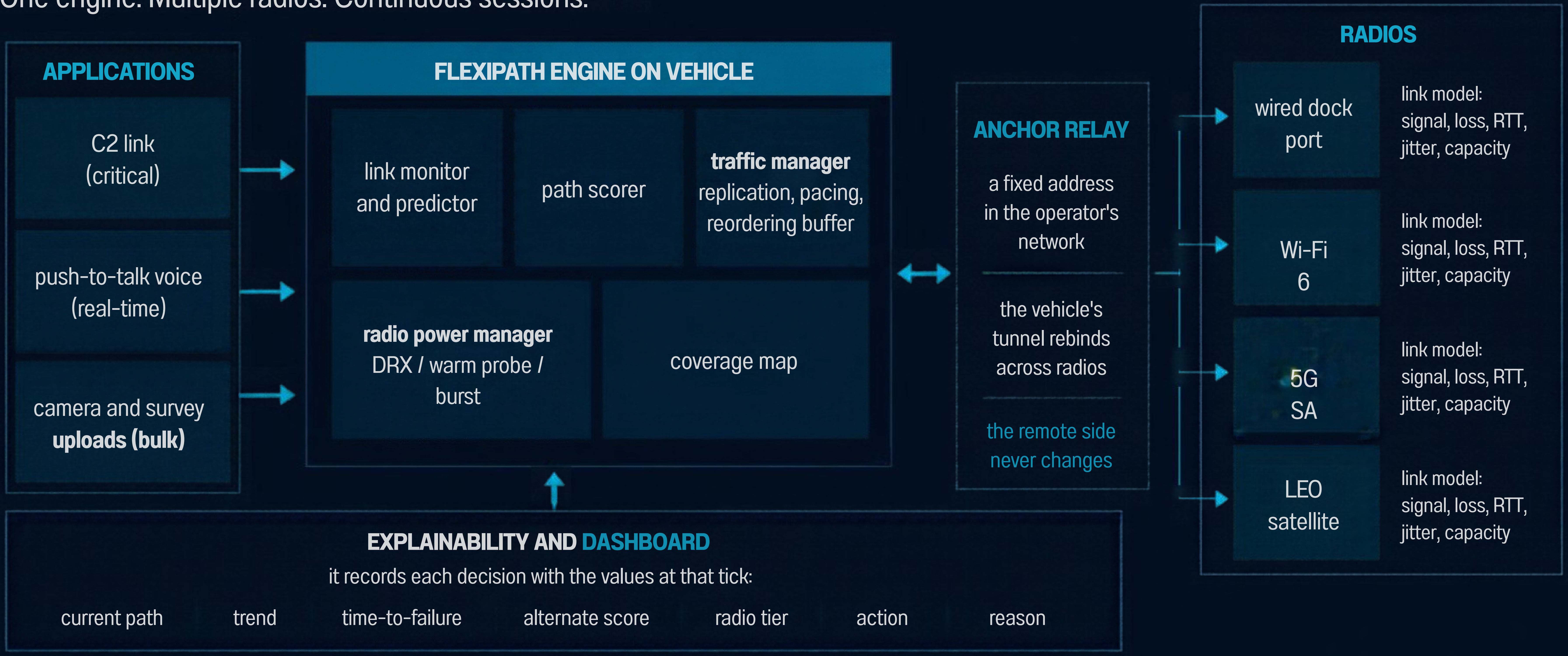
A reversed trend cancels the prediction at no cost.
If an outage comes with no warning, use a fast failover (160 ms in our model).

SWITCH and CONTINUE

Make the tested network the main one with a 0 ms gap. Put the old radio back to sleep.
Connections stay open in a tunnel to the anchor relay.

System design

One engine. Multiple radios. Continuous sessions.



APPLICATIONS

C2 link (critical)

push-to-talk voice (real-time)

camera and survey **uploads (bulk)**

FLEXIPATH ENGINE ON VEHICLE

link monitor and predictor

path scorer

traffic manager
replication, pacing, reordering buffer

radio power manager
DRX / warm probe / burst

coverage map

ANCHOR RELAY

a fixed address in the operator's network

the vehicle's tunnel rebinds across radios

the remote side never changes

RADIOS

wired dock port	link model: signal, loss, RTT, jitter, capacity
Wi-Fi 6	link model: signal, loss, RTT, jitter, capacity
5G SA	link model: signal, loss, RTT, jitter, capacity
LEO satellite	link model: signal, loss, RTT, jitter, capacity

EXPLAINABILITY AND DASHBOARD

it records each decision with the values at that tick:

current path · trend · time-to-failure · alternate score · radio tier · action · reason

DESIGNED TO SIT UNDER ANY COMMAND-AND-CONTROL APPLICATION

it needs no change to the application

Prototype, simulation and model (criterion 03)

flexipath.vercel.app

interactive 3D digital twin with mission and advanced views

guided demo

pauses at each real decision, explains what happened and which panel shows it

deterministic 10 Hz engine

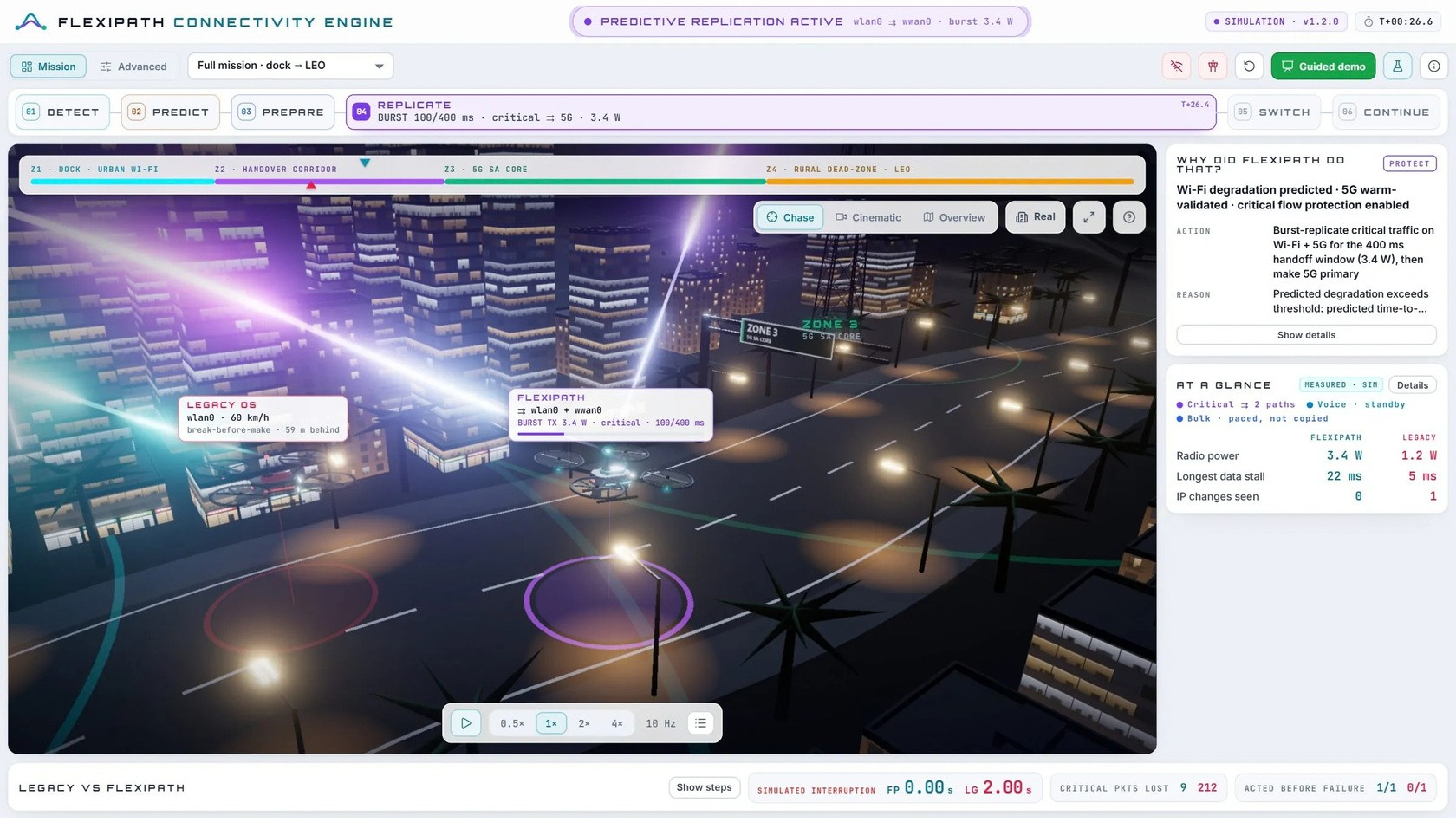
seeded random streams with fading, capacity, loss, RTT, jitter, TCP and radio models

headless benchmark

scenarios A–H × 5 seeds × 4 speeds = 160 paired runs per stack, plus baselines and ablations; CSV output with 95 % CIs

published model

we check parameters against published measurements and test the 100 ms modem wake at 260 ms as a sensitivity run


FLEXIPATH CONNECTIVITY ENGINE
PREDICTIVE REPLICATION ACTIVE wlan0 ⇉ wwan0 · burst 3.4 W
SIMULATION · v1.2.0
T+00:26.6
Mission
Advanced
Full mission · dock → LEO
Guided demo
01 DETECT
02 PREDICT
03 PREPARE
04 REPLICATE
BURST 100/400 ms · critical ⇉ 5G · 3.4 W
T+26.4
05 SWITCH
06 CONTINUE
Z1 · DOCK · URBAN WI-FI
Z2 · HANDOVER CORRIDOR
Z3 · 5G SA CORE
Z4 · RURAL DEAD-ZONE · LEO
Chase
Cinematic
Overview
Real
ZONE 3
ZONE 3
5G SA CORE
LEGACY OS
wlan0 · 60 km/h
break-before-make · 59 m behind
FLEXIPATH
⇉ wlan0 + wwan0
BURST TX 3.4 W · critical · 100/400 ms
0.5×
1×
2×
4×
10 Hz
WHY DID FLEXIPATH DO THAT?
PROTECT
Show details
AT A GLANCE
MEASURED · SIM
Details
● Critical ⇉ 2 paths
● Voice · standby
● Bulk · paced, not copied
LEGACY VS FLEXIPATH
Show steps
SIMULATED INTERRUPTION FP 0.00 s LG 2.00 s
CRITICAL PKTS LOST 9 212
ACTED BEFORE FAILURE 1/1 0/1

Wi-Fi degradation predicted · 5G warm-validated · critical flow protection enabled

ACTION	Burst-replicate critical traffic on Wi-Fi + 5G for the 400 ms handoff window (3.4 W), then make 5G primary
REASON	Predicted degradation exceeds threshold: predicted time-to-...

	FLEXIPATH	LEGACY
Radio power	3.4 W	1.2 W
Longest data stall	22 ms	5 ms
IP changes seen	0	1

Mission view of the live prototype. The FlexiPath drone (cyan) and the legacy-stack drone (crimson) fly with the same seed and radio conditions. The decision loop, decision card and scoreboard are around the twin.

▶️ Try it yourself: flexipath.vercel.app · the guided demo, the simulator and the 2-minute video

Results against the existing mechanisms (criterion 04)

Paired means over 20 runs per scenario, same seeds for every stack.

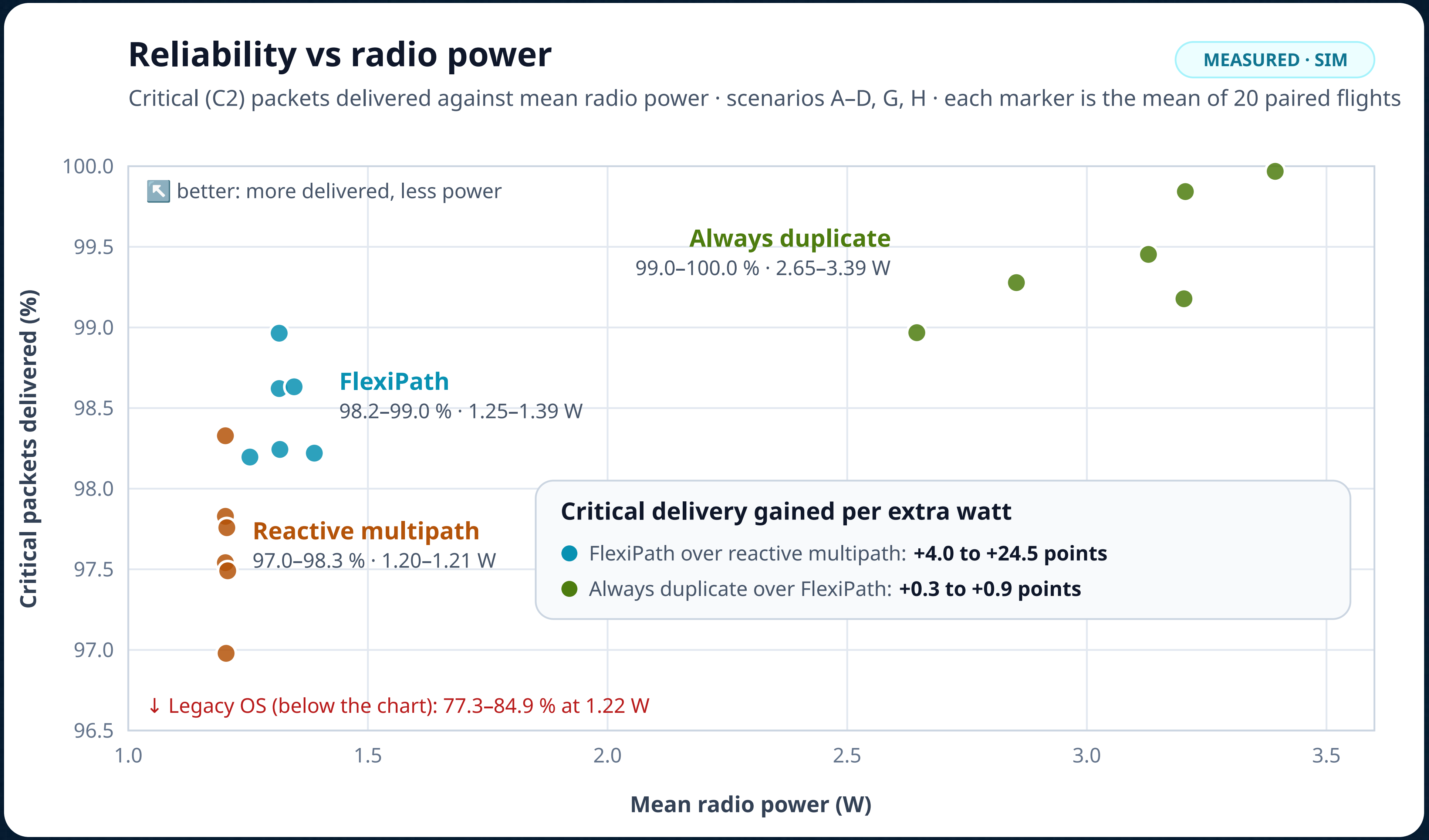
100% of flights where FlexiPath keeps C2 (flight count) under our piloted 300 ms rule

1.2–4.6 s FlexiPath acts before the path fails, versus 0.1–0.5 s after for reactive multipath

35% fewer critical packets lost vs reactive multipath (CI 30–39 %); held-out; ~20 % with equal detection

4.0–7.1 pp/W Reliability per extra watt versus reactive multipath **(16.9–24.5 in D)**

4.5–21× FlexiPath's extra watts give 4.5–21× more reliability than always duplicate's **0.3–0.9 pp/W**


Reliability vs radio power

MEASURED · SIM

Critical (C2) packets delivered against mean radio power · scenarios A–D, G, H · each marker is the mean of 20 paired flights

Critical delivery gained per extra watt

- ● FlexiPath over reactive multipath: **+4.0 to +24.5 points**
- ● Always duplicate over FlexiPath: **+0.3 to +0.9 points**

Reliability against radio power, scenarios A–D, G and H (paired means). FlexiPath is between the two existing designs. It gives 0.6–1.2 points more critical delivery than reactive multipath for 0.05–0.18 W more. It uses 2.1–2.6× less power than always duplicate.

Flights that keep C2 under the piloted 300 ms rule (Legacy · Reactive multipath · Always duplicate · FlexiPath): A, B, C 0 · 100 · 100 · 100 — D 0 · 0 · 100 · 100 — G, shared radio shadow 0 · 0 · 40 · 100 — H, map out of date 0 · 0 · 100 · 100.

Robustness: the ablations show trend prediction gives the gain; in our model, the burst adds no measurable reliability. The results hold on held-out seeds. Under the 1 s waypoint rule, every multipath stack keeps C2. So the piloted-rule advantage needs a spare modem that resumes fast.

Guidelines for QoS and QoE across heterogeneous networks (criterion 05)

01 Anchor the session, not the interface – a fixed **anchor relay** prevents every reset.

02 Act on the trend, not on the failure – a **3 s signal trend** gives seconds of warning.

03 Prepare the next path just in time, and only then – test it at **1.8 W**; do not keep radios on.

04 Duplicate only what needs it, only around the handover – a **400 ms burst** costs ≈ 1 % of protected bytes.

05 Deliver the **first copy**; never make real-time traffic wait in order.

06 Detect a failed path faster than the application allows – within the **300 ms C2** budget.

07 **Hold still** unless a move clearly helps – a reversed trend cancels the prediction at no cost.

08 Measure what **users feel**, on matched pairs: interruption, delivery, voice MOS, power, same seeds.

09 Use what you know about the route – a **coverage map** is better than reaction if two radios fail together.

Plan for real-life use, limits, and declaration

Deployment path, known limits, and our AI-use declaration

▶ TRY THE LIVE PROTOTYPE

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guided demo, simulator and the 2-minute video

DEPLOYMENT

A userspace agent on the vehicle's companion computer (Linux) controls modems through standard interfaces.
The anchor relay is at the operator's ground station or cloud edge.
It runs under current C2 and video applications; they need no change.

LIMITS

All results are simulated.
We evaluate the radio model at ground level.
Piloted rule needs a spare modem that resumes in ~**100** ms (5G RRC_INACTIVE or connected-mode DRX).
At **260 ms**, an unexpected outage still forces a hover.

REAL-LIFE USE

Phase 1 (bench): laptop with Wi-Fi and a 5G dongle runs the included handover probe to measure one real network change. **Phase 2** (field): test drone with Wi-Fi, 5G and a LEO terminal on a surveyed corridor; first survey flights build the coverage map. **Phase 3** (pilot): one operator, one corridor, the paired-comparison method of this study applied to real flights.

AI-USE DECLARATION

AI tools (Anthropic Claude) helped with code, analysis drafts, this deck and the video. An AI voice (Kokoro) reads the narration.
The design decisions, the experiments and their interpretation are the team's own, and every member can explain their part in their own words.

FlexiPath
5G